

# Linguistic Analysis of Referential Expressions and Lexical Variation in Discourse on Climate Data Visualizations

*Final Project Report — Data Science Project, Télécom Paris*

Team: Xisco Moncet • João Pedro Barreto • Naoures Abassi • Youssef Chebil • Lise Peguet

Project supervisor: Anne-Flore Cabouat

## 1 Introduction, Hypotheses, and Motivations

This project investigates how people linguistically refer to elements of a data visualization, and whether the way they talk changes when the visualization depicts a familiar, real-world territory versus an unfamiliar, fictional one. It is part of a broader research effort on the influence of prior knowledge on the interpretation of climate-related visualizations. The central question is whether prior knowledge of a depicted region leaves a measurable trace in spontaneous spoken discourse.

The experimental corpus comes from semi-structured interviews. Each participant viewed two maps showing average temperature data, one fictional map and one real map of France, completed four predetermined tasks per stimulus, and then answered eleven readability questions. Sessions were recorded and transcribed. Our team analyzed the resulting transcripts, working only with the participant's speech turns and aligning each turn with the question being answered and the experimental condition. One characteristic of the sample is worth flagging at the outset: the ten participants were a homogeneous group of highly educated experts, all male, all holding at least a master's degree, and mostly researchers or engineers in computer science, human-computer interaction, and related fields, with uniformly high visualization literacy and geographic familiarity. The study therefore probes how expert users behave, which limits the generalizability of the findings, but might also strengthen some interpretations since general audience readers might face even worse difficulties.

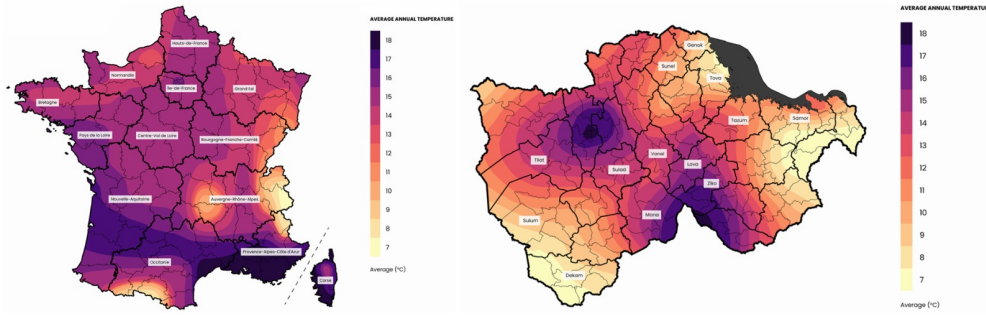
We organized the study around three hypotheses inherited from the project plan:

- **H1 (referential expressions).** Referential expressions used to describe the real, familiar map are more frequently specific and anchored to external knowledge, for example named regions and geographic landmarks, while expressions for the fictional map rely more on deictic or generic forms such as “this area” or “the previous one.”
- **H2 (lexical variation).** Lexical variation is higher in the real-map condition, reflecting richer semantic associations enabled by prior knowledge. This was to be measured through type-token ratio and related vocabulary-breadth metrics.
- **H3 (order effects).** The order of presentation, fictional first versus real first, modulates the discourse on the second stimulus, leaving measurable carry-over effects.

The motivation is both scientific and practical. Scientifically, if familiarity systematically changes referential strategy, then user studies that evaluate the readability of a visualization are partly measuring the reader's prior knowledge rather than the design itself, which has consequences for how such studies should be controlled. Practically, detecting a language bias linked to familiarity helps separate genuine design problems from effects that stem from what the viewer already knows.

## 2 Implementation Process

Our pipeline was deliberately simple and reproducible, built as a single Jupyter notebook that loads the transcripts from spreadsheet files, annotates each speech turn, computes the statistics for the three hypotheses, and exports a multi-sheet results workbook. The design followed six stages: verifying and correcting the transcripts, preprocessing and aligning them with the experimental metadata, defining an annotation scheme, extracting referential and lexical features, running the comparative statistics, and reporting.



**Figure 1.** Side-by-side presentation of the two temperature visualizations used in the study: the real map of France (left) and the fictional map (right).

## 2.1 Transcript verification and preprocessing

The transcripts were produced by automatic speech-recognition systems and arrived in heterogeneous quality: P1 to P6 were partially annotated and P7 to P10 were raw. The raw data contained inconsistent participant and condition labels, fragmented speaking turns split across rows, recognition errors, oral fillers, and a spurious “Speaker 3”. In practice, this label sometimes corresponded to the interviewee and sometimes to an interviewer. On a few occasions, a second interviewer was present in the room, but since interviewers were not expected to interact with one another, their utterances could safely be merged with those of the main interviewer. A substantial verification and normalization phase was therefore necessary before any analysis. We standardized speaker labels (Speaker 1 as interviewer and Speaker 2 as interviewee), manually reassigned every “Speaker 3” turn from context, aligned each utterance with its visualization condition (real or fictional) and with the interview question it answered using a numeric identifier from 1 to 13 (questions 1–11 corresponding to the map-related questions and identifiers 12 and 13 added by us for the two final concluding questions), and distinguished main questions from follow-ups. Consecutive utterances from the same speaker were merged into coherent turns, and transcription errors, especially geographic and fictional place names, were corrected by hand. We tested AI-assisted automatic correction on samples but found it semantically unreliable, so manual correction was preferred.

On top of the verified transcripts we built a reproducible cleaning pipeline in Python, applied as a chain of stages each preserved in its own column for transparency and debugging: `clean_text` (lowercasing, Unicode and apostrophe normalization, whitespace cleanup), then `no_punctuation_text`, then `no_fillers_text` (removing hesitation markers, conversational fillers, and discourse markers via a custom lexicon), then `no_repetition_text` (collapsing adjacent repeated words from disfluency). The cleaned corpus was exported as a spreadsheet, one row per turn, carrying the speaker, timing, condition, question number, question type, and every cleaning stage. We have an effective sample of ten participants and 995 interviewee turns, 506 for the real map and 489 for the fictional map.

An important design decision in the analysis code was to define one vocabulary list per referential category and reuse it everywhere. The same five lists drive both the H1 regular-expression annotation and the H2 semantic-field density measurement, which removes any risk of the two analyses silently diverging.

## 2.2 H1: annotation of referential expressions

We annotated five categories of referential or epistemic marker. GEO captures geographic proper names, both real French regions such as Bretagne or Corse and the invented names printed on the fictional map such as Tilat or Genok. DEF captures definite descriptions of map elements built as “the + noun,” for example “the scale” or “the region.” DEM captures demonstratives and pure spatial deixis such as “this one,” “here,” or “up here.” GEN captures generic or visual vocabulary such as “a region” or “some colours.” UNCERTAINTY captures epistemic hedges and difficulty markers such as “I think,” “not sure,” or “confusing.” Each turn received a count per category, and these counts were aggregated by condition, by participant, and by question. We report referential measures as rates per turn: the speaking turn is

| Label              | Description                              | Example expressions                                                      |
|--------------------|------------------------------------------|--------------------------------------------------------------------------|
| <b>GEO</b>         | Geographic proper names                  | <i>Bretagne, Corse, Paris, Tilat, Genok</i>                              |
| <b>DEF</b>         | Definite descriptions of map elements    | <i>the map, the scale, the region, the border, the temperature range</i> |
| <b>DEM</b>         | Demonstratives and spatial deixis        | <i>this region, that one, here, up here, over there</i>                  |
| <b>GEN</b>         | Generic visual and perceptual vocabulary | <i>gradient, contrast, dark, purple, shape</i>                           |
| <b>UNCERTAINTY</b> | Epistemic hedges and difficulty markers  | <i>I think, not sure, guess, confusing, struggling</i>                   |

**Table 1.** Referential and epistemic categories used in the annotation scheme, together with representative examples. Each speech turn received a count for each category, and these counts were subsequently aggregated by condition, participant, and question.

the natural unit of a referential act, and dividing by the number of turns normalizes for the fact that participants and conditions differ in how much speech they produce. The lexical-diversity metrics of H2 are the one exception and are computed per participant, since a type-token ratio is only meaningful over a sizeable token sample rather than over a single short turn.

### 2.3 H2: lexical variation

For lexical variation we concatenated all turns per condition and tokenized once with spaCy, lemmatizing and removing punctuation. We computed the raw type-token ratio, the segmented type-token ratio over forty-token windows to neutralize the length bias of raw TTR, the MTLT, and Yule’s K. We added a part-of-speech distribution and, as a complementary view, the density of each of the five shared vocabulary fields. To test the hypothesis properly we computed a per-participant type-token ratio in each condition and compared the paired values with a Wilcoxon signed-rank test.

### 2.4 Statistical modelling and extensions

Because every participant speaks in both conditions, the turns from one person are not independent. A plain regression would treat them as independent and could therefore bias significance estimates. We therefore fitted a linear mixed-effects model for each label, with condition as a fixed effect and a random intercept per participant, which is the statistically appropriate model for this repeated-measures design. Beyond the aggregate test we broke the data down three ways. We computed every label’s counts and rates per participant, which exposes individual variation and feeds the mixed model. On the recommendation of our supervisor, we grouped the eleven interview questions according to the four dimensions identified in the PREVis framework (Cabouat et al., 2026), whose structure emerged from an exploratory factor analysis in a large-scale study. These blocks correspond to clutter (questions 1 to 3), retrieval (questions 4 to 6), understanding dataset features (questions 7 and 8; the “DataFeat scale”), and higher-order evaluation (questions 9 to 11; the “Understand scale”). We then computed label rates per block to determine at which stages of processing referential strategies differ most.

We examined participant-to-participant similarity descriptively, using cosine similarity between each participant’s label profile, to confirm that the condition effect is broadly shared rather than driven by one outlier. We deliberately kept this similarity analysis descriptive: the hypotheses concern conditions, not which person talks differently, so inferential pairwise testing between participants would not be meaningful and would invite multiple-comparison problems.

For H3 the carry-over logic was implemented as a reusable module that runs on the same spreadsheet corpus as H1 and H2, so all three hypotheses share identical data. It runs four non-parametric tests, comparing the first and second stimulus within each participant and comparing the two order groups between participants, each reported with an effect size and bootstrap confidence intervals. Crucially the presentation order is inferred from the data, the first condition a participant actually discusses, rather than

## PREVis questionnaire

There are 4 individual scales in PREVis, each measuring a particular dimension of perceived readability.

### Understand scale

- obvious** It is **obvious** for me how to read this visualization
- represent** I can easily understand **how the data is represented** in this visualization
- understand** I can **easily understand** this visualization

### Layout scale

- messy** I **don't** find this visualization **messy**
- crowd** I **don't** find this visualization **crowded**
- distract** I **don't** find **distracting parts** in this visualization

### Reading data scale

- find** I can easily **find specific elements** in this visualization
- identify** I can easily **identify relevant information** in this visualization
- information** I can easily **retrieve information** from this visualization

### Reading features scale

- visible** I find data features (for example, a minimum, or an outlier, or a trend) **visible** in this visualization
- see** I can **clearly see** data features (for example, a minimum, or an outlier, or a trend) in this visualization

**Figure 2.** Overview of the four PREVis dimensions used to group the interview questions, derived from the PREVis framework and available on <https://aviz.fr/PREVis/>, these dimensions correspond to clutter, retrieval, understanding dataset features, and higher-order evaluation groups we used. Example items from each scale are included to clarify the types of questions and cognitive processes represented by each block.

trusted from the filename, which is what allowed us to detect the P9 labelling error.

On the supervisor’s suggestion we added three extensions. First, a PRIOR\_KNOWLEDGE tag, inspired by Arunkumar et al. (2025), whose external-knowledge / extrapolation recall category is reported as the most discriminative signal between familiar and unfamiliar charts. Where GEO captures naming a real place, PRIOR\_KNOWLEDGE captures reasoning from world knowledge, for example “I am used to it” or “usually warmer,” giving a second, inference-based channel for H1. We constructed the tag empirically, testing each candidate sub-field for real-versus-fictional discrimination and keeping only the explicit knowledge markers. Their use turns out to be  $1.73\times$  more frequent on the real map than on the fictional one — an observed result of the retained markers, not a preset selection threshold; topographic nouns such as “coast” and causal connectives such as “because” describe both maps and were dropped to avoid adding noise. Crucially the tag does not re-list place names, so it does not double-count with GEO. Second, we merged the post-task survey scores onto the corpus from the two participant surveys, which record, for every participant, self-reported geographic familiarity and visualization literacy on a 1-to-5 scale and a graded score out of seven on each of the two maps. We used these measures to re-test the GEO effect while controlling for individual ability, although the relative homogeneity of the participants made it unlikely that this adjustment would have a major impact. Third, we correlated each participant’s linguistic profile with the grade they obtained on each map, and we analyzed the task grades in their own right.

## 3 Challenges and Findings

### 3.1 Challenges

The first challenge was the data itself. The transcripts were unverified and inconsistent, which is why the verification and preprocessing phase described above was so substantial. Two source-level problems deserve mention. Our H3 analysis infers the presentation order from the data, namely which condition the participant actually discusses first, rather than trusting the filename label. This caught a concrete mislabelling: P9’s working file is tagged fictional-first, yet the participant discusses the real map first, and both the original filename and the participant-profile document indicate real-first. The data-driven inference treats P9 correctly, and the source file should be fixed to match. The second challenge was the small sample, ten participants, which limits statistical power and means several effects can only be reported as exploratory. We addressed this with within-subject comparisons, non-parametric tests, and effect sizes reported alongside p-values. A third challenge was conceptual hygiene: early versions of the code reused the same names for different concepts across H1 and H2, and two separate analyses were both informally called “H3” when one of them actually measured the real-versus-fictional contrast (that is, H1). We resolved this by adopting one shared vocabulary, reserving the H3 label for the carry-over effect alone, and reframing the anchoring analysis as evidence for H1.

### 3.2 H1 findings: one strong, robust effect

The headline result is that geographic naming is strongly and significantly tied to familiarity. Speakers name real regions more than twice as often when describing the real map, and this is the only H1 effect that survives the mixed model.

| Label       | Rate Real | Rate Fict. | Ratio R/F | Mixed $p$ | Verdict               |
|-------------|-----------|------------|-----------|-----------|-----------------------|
| GEO         | 0.362     | 0.163      | 2.22×     | 0.000     | Supported             |
| DEF         | 0.982     | 0.980      | 1.00×     | 0.985     | Stable (as predicted) |
| DEM         | 3.265     | 3.314      | 0.99×     | 0.741     | Not supported         |
| GEN         | 0.383     | 0.299      | 1.28×     | 0.054     | Approaches support    |
| UNCERTAINTY | 0.717     | 0.765      | 0.94×     | 0.613     | Not supported         |

The directional predictions for DEM and UNCERTAINTY were not confirmed. Although the effect for GEN did not reach conventional significance levels, it approached significance and showed a trend in the predicted direction; a larger sample would help determine whether this tendency reflects a genuine effect, and further iterations of the coding scheme could reduce annotation noise and provide a more sensitive measure of generic vocabulary use. Demonstratives and deixis are extremely frequent in both conditions and do not separate them, which is intuitive once one realizes that pointing at the screen is the default mode of reference regardless of whether the territory is known. Uncertainty vocabulary trends in mixed directions and does not reach significance. The PRIOR\_KNOWLEDGE extension provides a second, independent confirmation of the familiarity effect: explicit appeals to world knowledge occur at 0.319 per turn for the real map against 0.188 for the fictional map, a 1.70-fold difference, and the mixed model gives a significant positive coefficient ( $p = 0.019$ ), with the direction holding for six of the nine participants in that analysis (P8's annotation was recovered afterwards). Typical real-map quotes are “because in France I'm used to it” and “the difference is because I know France”; on the fictional map this channel is largely unavailable, exactly as the theory predicts. The covariate analysis confirms robustness: the GEO mixed-model coefficient is essentially unchanged when visualization literacy is added (from +0.232 to +0.234, both  $p < 0.0001$ ), so the effect is driven by territorial familiarity rather than general chart-reading skill. The interaction with prior geographic familiarity is positive, more knowledge giving a slightly larger GEO effect, but not significant at this sample size ( $p = 0.197$ ).

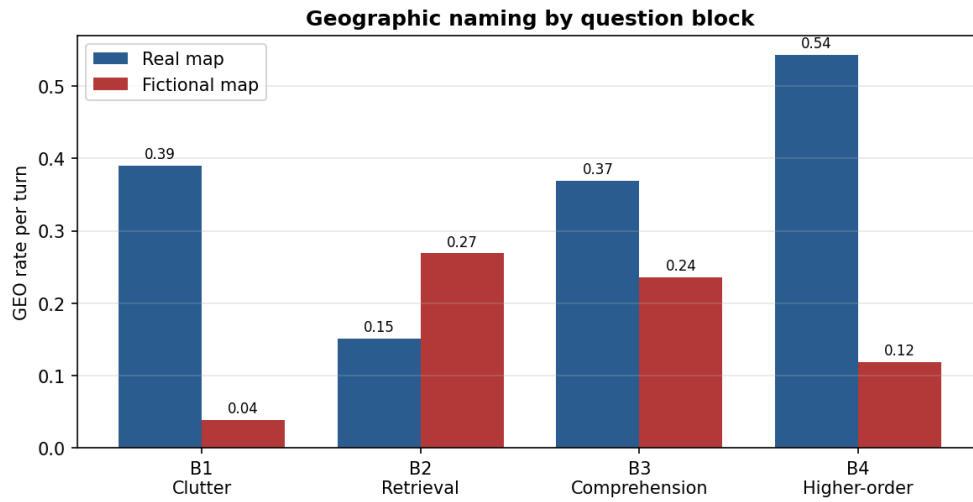
The question-block analysis sharpens the picture. Geographic naming for the real map is weakest during low-level retrieval tasks and peaks during higher-order evaluation and comparison, where the GEO rate for the real map reaches 0.543 per turn against 0.119 for the fictional map. In other words, prior knowledge is mobilized most when the task demands reasoning about the whole map rather than locating a single value.

Looking at the same effect participant by participant adds useful nuance. The real-map advantage in geographic naming holds for the clear majority of participants, but it is not universal: P3 names more regions on the fictional map than on the real one, and a few participants (P5, P6, P9) name few regions in either condition. This individual variation highlights the importance of the mixed-effects model. By treating each participant's baseline as a random intercept, it confirms that the average GEO effect is real even though one or two individuals run against it.

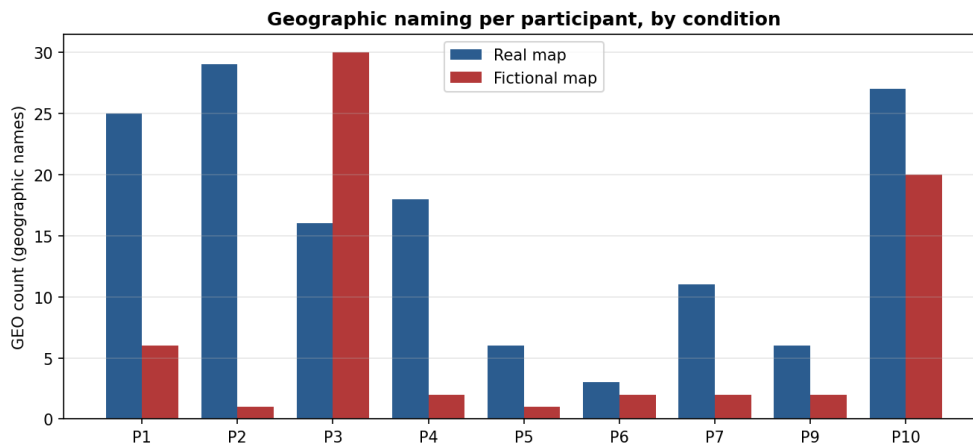
### 3.3 H2 findings: no diversity difference

H2 is not supported. The segmented type-token ratio is essentially identical between conditions (0.729 versus 0.728), MTLTLD differs only trivially, and Yule's K moves in the direction opposite to the prediction. The Wilcoxon test on per-participant type-token ratios is far from significant ( $p = 0.65$ ).

The interpretation is instructive rather than disappointing. Overall vocabulary diversity is the same because the two conditions achieve it differently: real-map speakers add variety through geographic names, while fictional-map speakers compensate with deictic and visual vocabulary. The part-of-speech distribution tells the same story from another angle. Nouns and pronouns are almost identical across



**Figure 3.** Geographic-naming rate per turn, by question block and condition. The real-map advantage grows from near zero in the clutter block to more than four-fold in the higher-order block.



**Figure 4.** Geographic-naming counts per participant. Most participants name more regions on the real map; P3 is a notable exception, and the mixed model accounts for this individual variation.

conditions, but the fictional map draws slightly more adjectives (7.81 against 7.19 per cent), consistent with describing unfamiliar shapes and colours, and slightly more adverbs, consistent with more hedging. The semantic-field densities, computed on the same labels as H1, show the substitution directly: geographic and generic-visual vocabulary lean towards the real map, while demonstrative and definite vocabulary lean towards the fictional one.

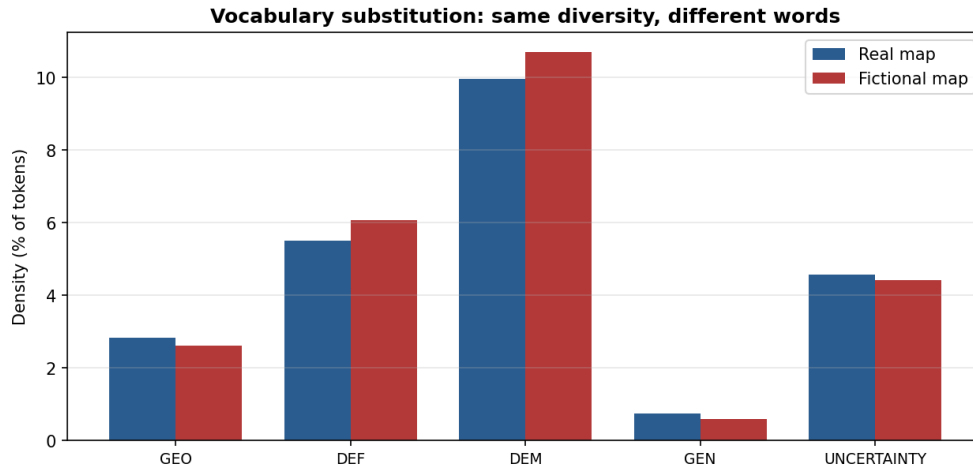
The per-participant view of lexical diversity confirms that no systematic gap exists. Some participants are slightly more diverse on the real map, others on the fictional map, and the differences are small and inconsistent, which is precisely why the paired Wilcoxon test returns a high p-value.

So the lexical analysis ends up reinforcing H1: the difference is in which words are used, not in how many distinct words are used.

### 3.4 H3 findings: within-subject carry-over

H3 is supported, but as a within-subject carry-over effect rather than a between-group order effect. When participants describe their second map, they make significantly more explicit references back to the first map and lean more heavily on deixis (context-dependent expressions such as “this one,” “here,” or “over there”), while their local vocabulary diversity falls. Four non-parametric tests, summarised in the table below, make this precise: each reports a *p*-value and an effect size, with the 95% bootstrap confidence intervals for every effect size provided in the accompanying results workbook.

| Metric                     | Real   | Fictional | $\Delta (R-F)$ | Supports H2?    |
|----------------------------|--------|-----------|----------------|-----------------|
| TTR                        | 0.054  | 0.052     | +0.002         | Neutral         |
| STTR (length-corrected)    | 0.729  | 0.728     | +0.001         | Neutral         |
| MTLD                       | 48.17  | 47.75     | +0.42          | Negligible      |
| Yule's K (lower = diverse) | 136.93 | 130.82    | +6.11          | Opposite        |
| Wilcoxon $p$ -value        | 0.652  | —         | —              | Not significant |



**Figure 5.** Semantic-field density (share of tokens) by condition, on the same five labels as H1. Geographic and generic vocabulary lean to the real map; demonstrative and definite vocabulary lean to the fictional map, so total diversity stays balanced.

We interpret these effects as evidence that describing the second map reactivates the first stimulus: participants explicitly compare back to the map they saw earlier and rely more on pointing, while producing slightly less varied vocabulary locally. The between-group order effect is not significant, which suggests the carry-over is roughly symmetric, going real-to-fictional and fictional-to-real produce comparable effects. With nine participants this remains exploratory, which is why effect sizes accompany every  $p$ -value, but the within-subject signal is consistent and is the clearest H3 result the data support. This also corrects an earlier, under-powered impression that H3 was merely weak: the order-group lexical comparison alone is flat, but the paired, within-subject carry-over tests are significant.

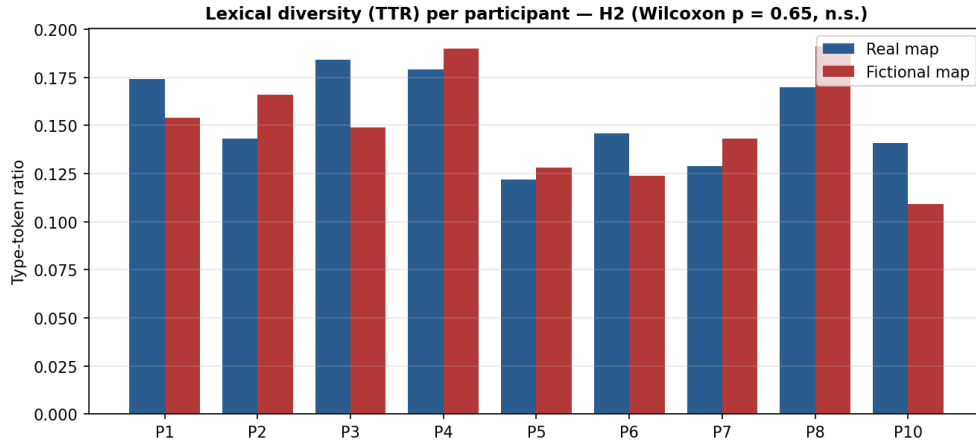
### 3.5 Task performance and what language does (and does not) predict

Having the graded survey results lets us connect the linguistic findings to how well participants actually performed the four tasks on each map. The grades, out of seven, tell a clear and somewhat counterintuitive story.

Note: There are four tasks per visualization (real and fictional), but this analysis uses a 7-point aggregated scale because of the subtask-level structure of R4 and N3, each composed of four subtasks (R4.1–R4.4 and N3.1–N3.4); the 7-point scale reflects the aggregated scoring across these subtasks.

Two things stand out. First, the fictional map was not harder than the real map; if anything participants scored higher on it. Once a reader understands how the colour-to-temperature encoding works, knowing the actual territory matters less than interpreting the visualization correctly. Second, there is a clear order effect in performance: participants who saw the real map first then scored markedly higher on the fictional map (an accuracy of about 86 per cent) than those who met the fictional map cold (about 63 per cent). Working through the familiar map first appears to teach the encoding, which then transfers to the unfamiliar one. This learning effect in the task scores is an independent corroboration, from behaviour rather than language, of the carry-over that H3 detected in the discourse: order matters.

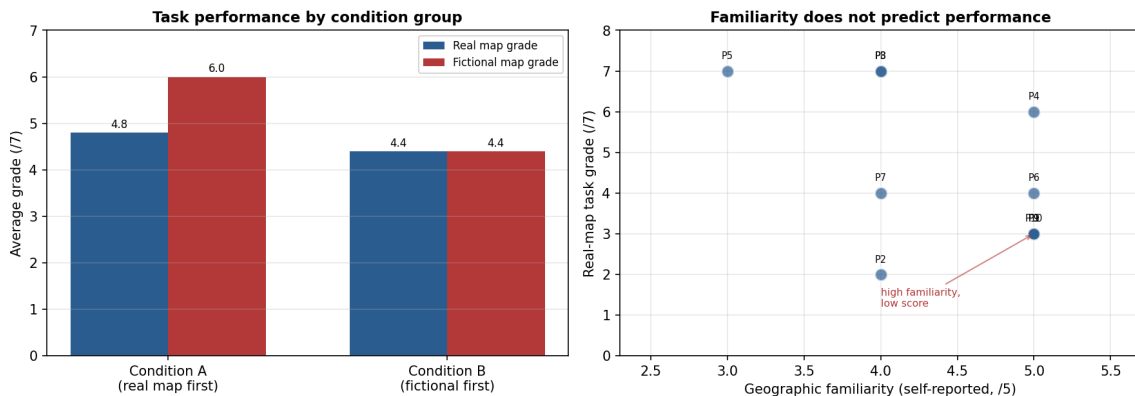
The most important link to our linguistic results is a dissociation. We correlated each participant's



**Figure 6.** Type-token ratio per participant in each condition. The direction of the small differences flips from participant to participant, and the paired Wilcoxon test is not significant ( $p = 0.65$ ), so H2 is not supported.

| Test      | Finding (first → second stimulus)         | $p$      | Effect size |
|-----------|-------------------------------------------|----------|-------------|
| T2 (core) | Cross-condition references rise           | 0.004    | $r = 0.44$  |
| T2        | Deictic references rise                   | 0.012    | $r = 0.37$  |
| T2        | Type-token ratio drops                    | 0.039    | $r = 0.25$  |
| T4        | Less deixis in the 5-turn transition zone | 0.027    | $r = 0.29$  |
| T1 / T3   | Order effect between groups               | $> 0.05$ | trend only  |

per-map feature rates with their grade on that map, across eighteen participant-map units, and found no significant relationship: uncertainty ( $\rho = -0.05$ ), prior-knowledge invocation ( $\rho = -0.03$ ), and geographic naming ( $\rho = -0.34$ ,  $p = 0.17$ ). The mild negative trend for geographic naming, together with the scatter on the right of Figure 5, where participants reporting maximal familiarity with France still scored only three out of seven on the real map, makes the point precisely. Familiarity drives how people describe a map, the anchored, region-naming style captured by GEO and PRIOR\_KNOWLEDGE, but it does not drive how accurately they read it. Task success is limited by visualization interpretation, not by territorial knowledge. The covariate models confirm the same separation from the other direction: the GEO effect is essentially unchanged when visualization literacy is added (coefficient  $+0.232$  to  $+0.234$ ), and indeed every participant in this expert sample rated their visualization literacy at four or five out of five, so literacy cannot explain the naming difference. Language reflects description strategy; performance reflects interpretation; the two are distinct.



**Figure 7.** Left: average task grade by condition group; the fictional map is not harder, and seeing the real map first lifts later fictional performance. Right: self-reported geographic familiarity against real-map grade; the two are unrelated, with several maximally familiar participants scoring poorly (a Dunning–Kruger-style pattern).

| Condition group              | Real-map grade | Fictional-map grade |
|------------------------------|----------------|---------------------|
| A, real map shown first      | 4.8 / 7        | 6.0 / 7             |
| B, fictional map shown first | 4.4 / 7        | 4.4 / 7             |

## 4 Alignment With the Proposed Plan

The project followed the original plan closely in structure while diverging in two deliberate ways. The plan’s pipeline, verify transcripts, preprocess and align, choose an annotation method, extract features, run comparative statistics, and report, was followed end to end. The risk analysis in the plan anticipated exactly the problems we met: limited data, transcript-quality issues, and possible condition imbalance. Our mitigations matched what we had promised, namely within-subject designs, non-parametric tests, effect sizes, and explicit handling of order.

The first deliberate divergence concerns method. The plan proposed comparing rule-based annotation, spaCy, and CamemBERT, with few-shot LLM annotation where relevant. In practice the interviews were conducted and transcribed in English, so CamemBERT, a French model, was unnecessary. A transparent rule-based annotation built on a shared vocabulary, combined with spaCy for tokenization and part-of-speech tagging, proved both sufficient and far more interpretable than a fine-tuned model would have been on so few data. The second divergence is that we went beyond the plan by adding the mixed-effects model, the PRIOR\_KNOWLEDGE construct, the survey-score covariates, and the task-performance correlation, all of which strengthened the H1 story and addressed the supervisor’s feedback.

## 5 Did We Accomplish What We Set Out to Do?

Largely yes, with honest qualifications. We delivered a complete, reproducible pipeline and tested all three hypotheses. H1 produced a clear, statistically robust result for geographic naming, which is the most important and most defensible finding of the project, and it was independently corroborated by the PRIOR\_KNOWLEDGE channel and shown to be robust to visualization-literacy covariates. H2 was tested rigorously and confidently answered in the negative, which is a genuine result rather than a failure: prior knowledge changes the kind of vocabulary, not its diversity. H3 turned out better than an early, under-powered look suggested: the carry-over effect is significant within subjects, even though the between-group order effect is not. Across the three hypotheses we therefore have one strong positive (H1), one clear negative (H2), and one supported-but-exploratory positive (H3).

The main thing we could not fully achieve was statistical power. The mixed-model and carry-over analyses were run on the nine participants available at the time (P8’s condition annotation was recovered later, bringing the corpus to ten), and even so only the largest effects clear the significance bar comfortably; the marginal generic-vocabulary effect ( $p = 0.054$ ) and the familiarity interaction ( $p = 0.197$ ) are suggestive but cannot be claimed. The task-performance analysis, at ten participants and five per condition, is likewise exploratory, and we report it as such. What we gained by bringing in the graded surveys, however, is a convergence of three independent signals, the linguistic GEO effect, the behavioural learning effect, and the familiarity-performance dissociation, that together tell a coherent story. We were transparent about the uncertainty throughout and report effect sizes alongside every p-value.

## 6 What We Would Do Differently, and Lessons Learned

If we started again, the single biggest change would be to invest in data quality and metadata first, before any analysis. Much of our effort went into reconciling inconsistent transcripts and an unreliable order label. Agreeing on a strict, documented transcript format and verifying the experimental metadata at the very start would have saved time and prevented the kind of problems our pipeline had to detect after the fact: P8 reaching us with no condition annotation, and P9’s order label contradicting the data. Both should be fixed at the source so the corpus is clean from the start.

A second change would be to fix the annotation scheme and inter-annotator agreement earlier. We converged on a clean five-label scheme, but only after several iterations in which the same names meant different things in different places. Defining the categories once, with explicit examples and a measured agreement score on a shared subset, would have made the whole pipeline more defensible. Third, we would frame the analysis around a smaller number of pre-registered, well-powered tests rather than a broad sweep; with limited data, depth on the GEO effect is worth more than breadth across weak effects.

The lessons are concrete. Reusing a single source of truth, here one vocabulary, eliminates a whole class of silent inconsistencies. The right statistical model matters: moving to a mixed model changed which effects we could honestly claim. A negative result, properly tested, is informative and worth reporting clearly. And inferring structure from the data rather than trusting labels, as we did for presentation order, can catch real errors. Above all, with a small sample, intellectual honesty about uncertainty is more valuable than a longer list of significant-looking findings.

## 7 Team Contributions

Every member contributed to a distinct, critical part of the project, with redundancy on the most important paths.

- **Naoures Abassi:** preprocessing, data handling, task analysis, and project management. She coordinated the team, owned the transcript verification and alignment with the experimental metadata, and led the analysis of the experimental tasks.
- **João Pedro Barreto:** labeling, the H1 analysis, the mixed-effects model, and gathering and unifying the code. He defined the referential annotation scheme, implemented the H1 statistics, built the repeated-measures mixed model, and consolidated the separate analyses into a single notebook and results workbook.
- **Lise Peguet:** the H2 analysis, preprocessing, and labeling. She implemented the lexical-variation metrics (TTR, STTR, MTLT, Yule's K) and the Wilcoxon test, and contributed to transcript preprocessing and annotation.
- **Youssef Chebil:** the H3 analysis. He designed and implemented the carry-over and order-effect tests, including the logic that infers presentation order from the data and the suite of non-parametric tests with effect sizes.
- **Xisco Moncet:** task tests and unifying the code. He worked on the predetermined-task analysis and contributed substantially to integrating the team's code into one coherent pipeline. He also merged the H3 code onto the shared corpus and implemented the two supervisor extensions, the PRIOR\_KNOWLEDGE tag and the survey-score covariate analysis.

## 8 Teamwork and Tools

We worked in two complementary sub-groups, one focused on referential expressions (H1 and H3) and one on lexical variation (H2), while sharing a single codebase so the analyses stayed consistent. We held a weekly team sync and met with our supervisor roughly every two weeks. Code was version-controlled in a shared Git repository, the analysis lived in Jupyter notebooks, and results were exchanged as a multi-sheet Excel workbook that any member could open without running the code. The interview transcripts and survey scores were shared as spreadsheets, and we communicated day to day through a group chat. This light structure suited a small team and kept the single integrated notebook as the project's source of truth.

## References

- Arunkumar, A., et al. (2025). *Prior knowledge and external-knowledge recall in the interpretation of data visualizations*. (Used as the conceptual basis for the PRIOR\_KNOWLEDGE annotation:

the external-knowledge / extrapolation recall category is reported as the most discriminative signal between familiar and unfamiliar charts.)

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- Honnibal, M., & Montani, I. *spaCy: Industrial-strength natural language processing*. (Tokenization, lemmatization, and part-of-speech tagging.)
- Cabouat, A.-F., He, T., Isenberg, P., and Isenberg, T. (2026). *PREVis: Perceived Readability Evaluation for Visualizations*. *IEEE Transactions on Visualization and Computer Graphics*, 31. doi:10.1109/TVCG.2024.3456318. (Used as the basis for grouping the interview questions into four cognitive blocks. These dimensions (clutter, retrieval, understanding dataset features, and higher-order evaluation) originate from the PREVis framework and emerged from an exploratory factor analysis conducted in a large-scale study.)

## Appendix: Use of AI Tools

We used an AI assistant (Anthropic’s Claude) as a coding and writing aid throughout the project, and we want to be transparent about how. The assistant helped us refactor and unify our analysis code, in particular consolidating several separate notebooks into one pipeline, converting hard-coded transcript data to a clean spreadsheet-loading workflow, and removing duplicated vocabulary definitions. It helped implement and debug standard components: the spaCy preprocessing, the lexical-diversity metrics, the mixed-effects model setup, and the formatting of the multi-sheet results workbook. It also assisted in drafting and tightening this report.

All scientific decisions remained ours. The choice of hypotheses, the annotation categories and their vocabularies, the decision to use a mixed model, the interpretation of every result, and the honest framing of negative and exploratory findings were made and verified by the team. We reviewed all AI-assisted code and checked the reported figures against the actual outputs of our notebook. The AI did not generate experimental data, run the experiment, or decide what the results mean; it accelerated implementation and exposition while the analysis and conclusions are the team’s own.